

Python Code

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3  #Funcionando ok!
4  class De:
5      def __init__(self, fitness_func, inf_limit, sup_limit, pop_size, genes,
6      MUT_FACTOR=0.4, custo=True, GERACOES = 300):
7
8          self.pop = np.random.uniform(low=inf_limit, high=sup_limit,
9          size=(pop_size, genes))
10         self.fitness = fitness_func
11         self.pop_size = pop_size
12         self.fitness_por_geracao = np.zeros(shape=GERACOES, dtype=float)
13         self.geracoes = np.linspace(start=1, stop=GERACOES, num=GERACOES,
14         dtype=int)
15
16         for j in range(GERACOES):
17             print(j)
18             new_pop = np.zeros((pop_size, genes), float)
19             for i in range(pop_size):
20                 first = self.pop[np.random.randint(pop_size)]
21                 second = self.pop[np.random.randint(pop_size)]
22                 third = self.pop[np.random.randint(pop_size)]
23                 new_pop[i] = first + MUT_FACTOR * (second - third)
24                 if custo:
25                     if fitness_func(self.pop[i]) > self.fitness(new_pop[i]):
26                         self.pop[i] = new_pop[i]
27                 else:
28                     if fitness_func(new_pop[i]) > self.fitness(self.pop[i]):
29                         self.pop[i] = new_pop[i]
30
31             fitness_arr = np.array([fitness_func(ind) for ind in self.pop])
32             melhor_idx = np.argmax(fitness_arr)
33             self.fitness_por_geracao[j] = fitness_arr[melhor_idx]
34
35     def best(self):
36         self.best = self.pop[0]
37
38         for i in range(self.pop_size):
39             if self.fitness(self.pop[i]) < self.fitness(self.best):
40                 self.best = self.pop[i]
41
42         return self.best
43
44     def plot(self):
45         fig, ax = plt.subplots()
46
47         ax.plot(self.geracoes, self.fitness_por_geracao)
48
49         ax.set_xlabel("Geração")
50         ax.set_ylabel("Fitness")
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48     ax.set_title("Algoritmo DE")
49     ax.legend(["Fitness"])
50
51     plt.show()
52
```